



06/20/20

1. Summary:

- 1.1. *The purpose of this document is to gain a better understanding of a class management application called Oswald, to see if it has the potential to expand to other classes. Right now this is an unknown, but if the right customers are approached to understand how it works, the initial efforts could lead to a greater return.*
- 1.2. *Oswald is a full-stack enterprise application used for automating the entire process of running a class. Daniel Izadnegahdar created Oswald in 2017 and it took 3 years and 50k lines to complete. The work it currently does equates to about \$25,000 a year in teaching assistant costs for the 2 engineering classes that Dan teaches at CSU.*
- 1.3. *The business model of this proposal is similar to what Oracle or SAP is doing with businesses to build custom enterprise applications. Only this one is more geared towards classes, using the Oswald blueprint, and at a much lower licensing cost to Oracle or SAP.*

2. Market Target:

- 2.1. *Oswald can target any class at any education level i.e. k to 12, community colleges, university courses, or training seminars. The 3 main efforts in running any class are the following: (1) lecturing, (2) scheduling, and (3) grading. Oswald is designed to automate the last 2 items. Any class that has homework, exams, and a schedule can be automated with Oswald.*
- 2.2. *The paying customer would be an e-learning specialist who manages the technology licenses of a school, but the app would benefit the teachers themselves.*

3. Market Evidence:

- 3.1. *In 06/14/17, Dan spoke with a series of 4th grade teachers who said "yes there are many small time software packages that can automate this and that, but nothing that integrates an all-in-one solution where I can press a button and export the final grade sheet at the end of the semester. I'm sure every teacher I know would love to have something like that".*
- 3.2. *On 08/21/19, Dan was invited to a teaching conference that included hundreds of professors at CSU who teach gateway courses. Gateway courses are the largest freshmen courses at universities and are high stakes because they are a key factor in determining student graduation. State funding is often determined by a school's graduation rate. A re-occurring topic in this conference was the lack of technology to automate grading, and several professors raised their hand to say "is there some sort of software that can reduce the teaching administration load?"*

3.3. Remote autonomous e-learning has been growing ever since the 2020 COVID lockdown. A lot of teachers have said that remote learning has actually increased their workload instead of reduce it.

4. Cost:

4.1. An Oswald license would be a re-occurring license every semester and its cost would be determined by the amount of savings it provides a school. Below are 3 possible sources:

4.1.1. Reduction in teaching assistant administration costs by 85%.

4.1.2. Savings from mainstream licenses like Blackboard.

4.1.3. Improved quality of education with faster feedback and consistency.

5. Similar Products and Niche:

5.1. Blackboard:

5.1.1. Summary:

5.1.1.1. Blackboard is a \$700 million Washington based 3k employee company that is the mainstream equivalent of Oswald. However, Blackboard is expensive and designed to be a universal customizable tool, it requires time to setup every semester, and isn't as autonomous as Oswald. Blackboard also only grades multiple-choice and data entry. It isn't designed to grade cartesian or pdf drawings. Blackboard can also appear complicated to some students, Oswald is designed to be simple, with only a 'submission' and 'grading' page.

5.1.2. Visual Comparison:

Blackboard:

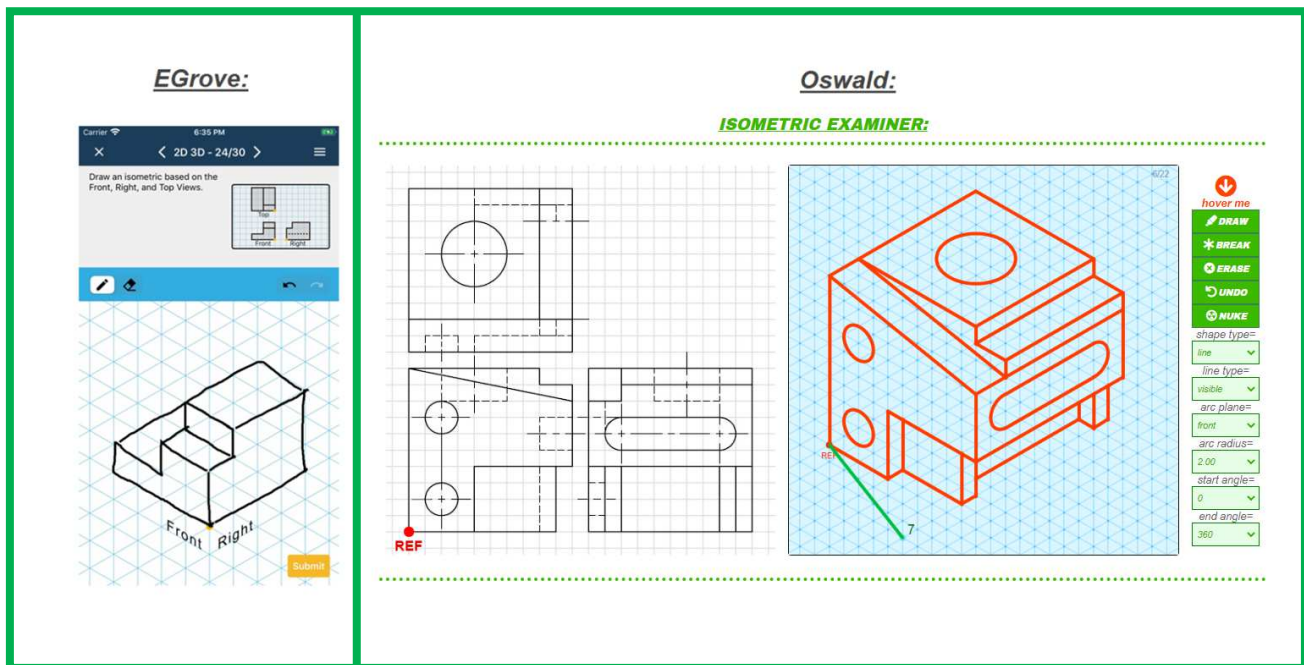
Oswald:

5.2. EGrove Education:

5.2.1. Summary:

5.2.1.1. EGrove Education Inc is a San Diego based 7-person 3-year old startup company that sells a Spacial Viz App as a downloadable widget on google play, to grade cartesian drawings. Oswald already had this function integrated, but the niche with Oswald is that it can integrate this data with the rest of the class data. Spacial Viz is just a standalone widget that teachers have to integrate manually with their master grade sheet.

5.2.2. Visual Comparison:



6. Benefits:

6.1. Simplicity:

6.1.1. Oswald is self-intuitive and is easy to learn.

6.2. Real-Time Feedback:

6.2.1. All of the grading is done by Oswald so students can check their grades at any time and see how they are doing compared to other students. A lot of assignments on Blackboard need to be graded by hand and causes delays.

6.3. All-In-One:

6.3.1. All of the data (accounts, schedules, submissions, grades, group projects) communicate with each other, everything is interconnected, and data updates together. This allows the app to generate a final excel sheet that provides an enormous convenience to teachers.

6.4. Fully-Autonomous:

6.4.1. All of the scheduling and grading tasks are done automatically by the application. Some of the smaller features it includes are auto-grouping, contribution voting with revenge protection, late penalty calculations, file validation with text-to-speech, variation randomizers to mitigate cheating, and OCR analysis.

7. Risks:

7.1. One-Off:

7.1.1. Oswald only works for 2 classes at CSU, and is highly 'hard-coded'. To make it work for any other class, a software architect would need to sit down with each professor, and specify how they want to run the class as far as assignments, exams, types of questions, and policies are concerned. Once this is done, the developer would need to update the app for it to work for that class specifically, which could take 2-3 months depending on the complexity. Once the setup is complete however, the app will run autonomously with very little effort for as many semesters as needed.

7.1.2. Another approach is to build generic versions of each class like 'chemistry' or 'physics' and let teachers adopt these for their class. Whether teachers would adopt these would depend on their preferences and how 'picky' they are with their class. In academics, teachers usually decide how they want to run their class; this is rarely dictated by school management.

7.2. Training:

7.2.1. Over the course of the semester, students are going to complain about their grades, and the teacher will need to know how to use Oswald to make custom changes. Historically, students have rarely needed training on how to use the student pages, but the teacher/administration pages would require training because they are more complicated.

7.2.2. Student Pages:

STUDENT ONLINE			
LOG	ACCOUNT	CONTENT	EXTRA CREDIT
LOGIN	SUBMISSIONS	ARCHIVE	THE FUN LABS
LOGOUT	GRADES	VIDEOS	BUILD YOUR OWN
	SETTINGS		
	GROUPS		

7.2.3. Teacher Pages:

MASTER ONLINE				
LOG	PROJECT	STUDENT	COURSE	TABLES
LOGIN	PRINT PROJECTS	BUILD STUDENTS	BUILD SCHEDULE	VIEW TABLES
LOGOUT	GRADE PROJECTS	VIEW STUDENTS	UPDATE SCHEDULE	EXPORT TABLES
		UPDATE ONE	UPDATE DEADLINE	QUERY BUILDER
		UPDATE MANY	RESET COURSE	
		RESET STUDENTS		

7.3. Database Architecture:

7.3.1. The database is currently designed to run 2 specific classes only. To run more classes, a database schema review and re-design would be required. Several design questions about licensing, data segregation, security, and data performance/scaling would need to be reviewed.

7.4. Support:

7.4.1. During the 1st year Oswald was released, releases were happening every 2 weeks with a ticket count of 30-40 items. Two years later, releases were quarterly with a ticket count of 1-2 items. Support will be required during the initial release of Oswald for new classes.

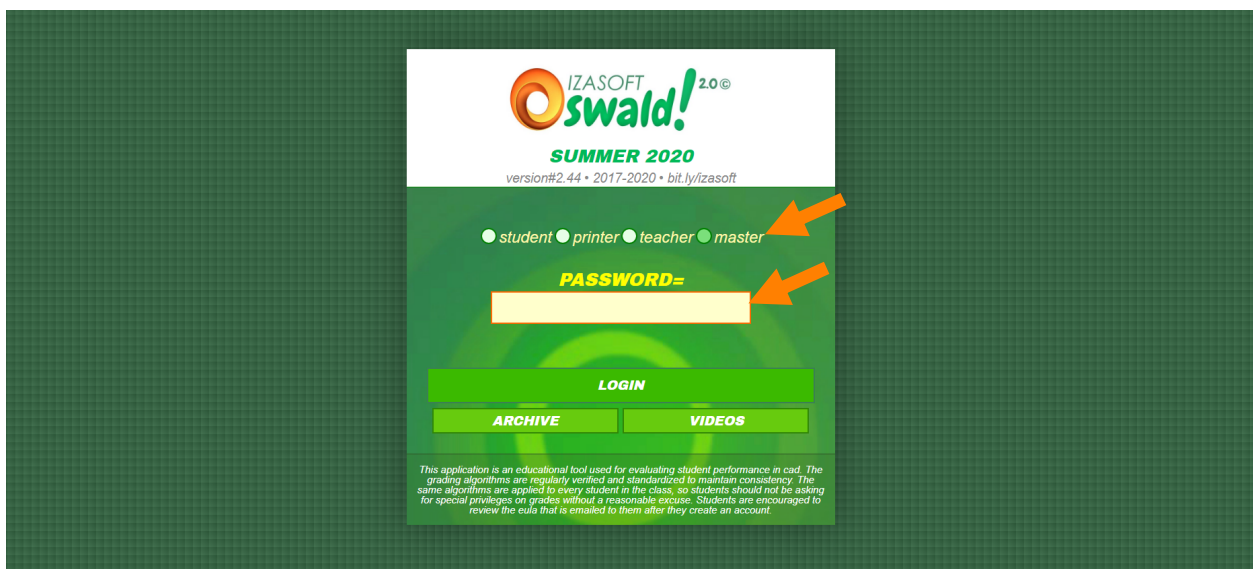
8. Semester Process:

8.1. Summary:

8.1.1. Every semester, the following steps need to occur for Oswald to run the class successfully.

8.2. Teacher:

8.2.1. Teacher logs in as master.



8.2.2. Teacher resets the previous schedules and students. Oswald runs on a semester by semester basis. Previous semester data is not stored in the database but teachers can export spreadsheets of each table if they want to keep a record.

.....
this will delete very session inside the schedule, semester, and deadline tables
(it will also clear the autoGroup/autoDelete dateStamps)=

DELETE ALL SCHEDULE SESSIONS

.....
this will delete every student account on the database

DELETE ALL ACCOUNTS

8.2.3. Teacher builds schedule.

BUILD SCHEDULE:

set session=

semester=

class code= ☒ mce180 ☐ mce181

next available session=

class location=

class start time (hh-mm-ss)= ☐ am ☒ pm

class end time (hh-mm-ss)= ☐ am ☒ pm

exact first day of class (mm-dd-yy)=

exact last day of class (mm-dd-yy) (should be finals week)=

weekly frequency= ☒ once ☐ twice

add holiday=

holiday name=

holiday date(mm-dd-yy)=

actions=

holiday list=

BUILD CUSTOM SCHEDULE

BUILD 180 SAMPLE

BUILD 181 SAMPLE

8.3. Student:

8.3.1. Student creates account selecting their schedule.

CREATE ACCOUNT:



firstName=

lastName=

phone#=

studentid#=

re-type studentid#=

email=

email code=

class= ☒ mce180 ☐ mce181

session= ☒ a

session details=

- A: mon/wed[04:10:00 PM - 05:45:00 PM]

password=

PRESS [SEND EMAIL CODE] BUTTON

RETURN TO SPLASH SCREEN

8.3.2. Student logs in:

8.3.3. Student views dashboard to see schedule. Student can check 'main' and 'side' columns to check codes for appropriate slide deck and youtube video provided on google.

DASHBOARD:
TODAY IS FRIDAY, JUNE 12, 2020!

CLASS REQUIRED!

sequence	location	date	startTime	endTime	submission	main	side	topic	teacher
01	remote	5/18/2020	04:10:00 PM	05:45:00 PM	n/a	ce01	sp01	orthographic projection	dan
02	remote	5/20/2020	04:10:00 PM	05:45:00 PM	n/a	ce02	sp38	isometric projection	dan
03	remote	5/25/2020	04:10:00 PM	05:45:00 PM	n/a	n/a	n/a	memorial day [no class]	dan
04	remote	5/27/2020	04:10:00 PM	05:45:00 PM	hw01	ce03	sp13 + sp22	sectioning	dan
05	remote	6/1/2020	04:10:00 PM	05:45:00 PM	hw02	ce04	n/a	general dimensioning	dan
06	remote	6/3/2020	04:10:00 PM	05:45:00 PM	n/a	ce05	n/a	hole dimensioning	dan
07	remote	6/8/2020	04:10:00 PM	05:45:00 PM	hw03	ce06	sp14	tolerancing	dan
08	remote	6/10/2020	04:10:00 PM	05:45:00 PM	n/a	sg01	sp28	exam review	dan
09	remote	6/15/2020	04:10:00 PM	05:45:00 PM	me01	n/a	n/a	midterm exam	dan
10	remote	6/17/2020	04:10:00 PM	05:45:00 PM	n/a	ce07	sp37	features i	dan
11	remote	6/22/2020	04:10:00 PM	05:45:00 PM	n/a	ce08	sp11	features ii	dan
12	remote	6/24/2020	04:10:00 PM	05:45:00 PM	n/a	ce09	sp06 + sp33	part drawings	dan
13	remote	6/29/2020	04:10:00 PM	05:45:00 PM	hw04	ce10	sp21 + sp39	assemblies	dan
14	remote	7/1/2020	04:10:00 PM	05:45:00 PM	n/a	ce11	sp07 + sp08	assembly drawings	dan
15	remote	7/6/2020	04:10:00 PM	05:45:00 PM	hw05	sg02 + ce12	sp26 + sp32	exam review + patterns + configurations	dan
16	remote	7/9/2020	04:10:00 PM	05:45:00 PM	fe01	fe01	n/a	final exam	dan

ASSIGNMENTS REQUIRED!

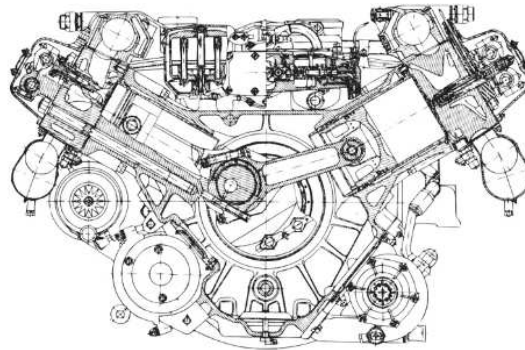
submitted assignments=	total assignments=	% completion=
3	7	42.86
you have to submit every assignment before this deadline=		
Friday, July 10, 2020 [5:45 PM]		
days left=	hours left=	minutes left=
28.23	677.56	40653.57

schedule

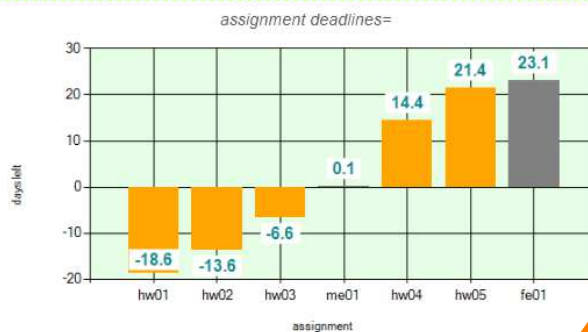
submission
summary

8.3.4. Student goes to the submission page, completes an assignment, and submits.

SUBMISSIONS:



when this assignment code changes, different examiners below will display, it can range from 1 - 6 examiners, and the examiners can be multiple-choice, file uploads, data-entries, or cartesian drawings



green= submitted, orange= accessible but not submitted, gray= not yet accessible

assignment header	
assignment attribute	value
code=	hw01
title=	homework#01
sub title=	orthographic projection
access date=	5/17/2020 11:59:59 PM
due date=	5/27/2020 11:59:59 PM
days left=	-18.62
start files=	errorcards.xlsx

validators control for proper submissions, there are up to 3 validators for each examiner

examiner#01			
examiner	examiner completion	%	validator(s)
ORTHOGRAPHIC		0	completion= ✖

once all examiners are complete, student clicks the submit button, where files are sent to archives, submission is graded, and database is updated.

YOU CAN ONLY SUBMIT ONCE!

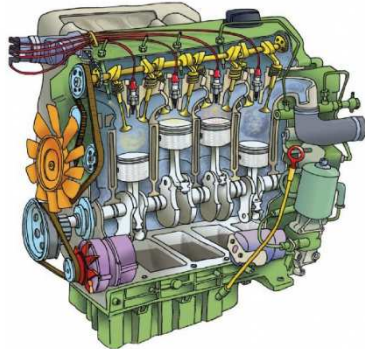
SUBMIT

Before submitting, make sure to 01) open your file one more time, and double check that it's exactly the way you want it because you can only submit once, 02) check the errorCards.xlsx file so you know exactly what the staff will be grading for, 03) keep a copy of your submission for your record, 04) do your own work and don't submit files from previous semesters because this is easy to detect and will flag your file. After clicking [SUBMIT], if a green confirmation stamp (at the bottom of the page) doesn't show up, it means your submission never submitted. Please review this submission stamp after submitting. Be patient with large files, they will take longer to submit.

✖ ASSIGNMENT WAS NEVER SUBMITTED!

8.3.5. Exactly 48 hours later, student can check back to see their grades on the 'grades' page.

GRADES:



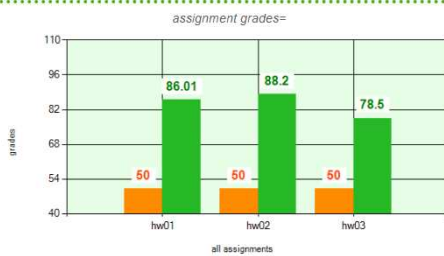
MASTER BREAKDOWN:
YOUR CURRENT GRADE IS AN **F** AS OF **HW03**
YOU ARE RANKED# **10** OUT OF **10** STUDENTS



green= all ranks, orange= your rank

DETAIL BREAKDOWN:

grade with curve(letter)= **F** **INFO**
grade with curve(numerical)= **50** **INFO**
curve(numerical)= **0** **INFO**
grade(numerical)= **50** **INFO**



green= average grade, orange= your grade
the classAverageGrade does not include no submissions

assignment	status	assignment grade	examiner#01, 02, 03 grade				file#01, 02, 03 grade		late penalty	offset(s)
hw01	not submitted!	50 INFO	50 INFO	NULL	NULL	NULL	NULL	NULL	50 INFO	0 INFO
hw02	not submitted!	50 INFO	50 INFO	NULL	NULL	NULL	NULL	NULL	50 INFO	0 INFO
hw03	not submitted!	50 INFO	50 INFO	NULL	NULL	50 INFO	NULL	NULL	50 INFO	0 INFO
hw04	not submitted!	NULL INFO	NULL INFO	NULL	NULL	NULL	NULL	NULL	NULL INFO	NULL INFO
hw05	not submitted!	NULL INFO	NULL INFO	NULL	NULL	NULL	NULL	NULL	NULL INFO	NULL INFO

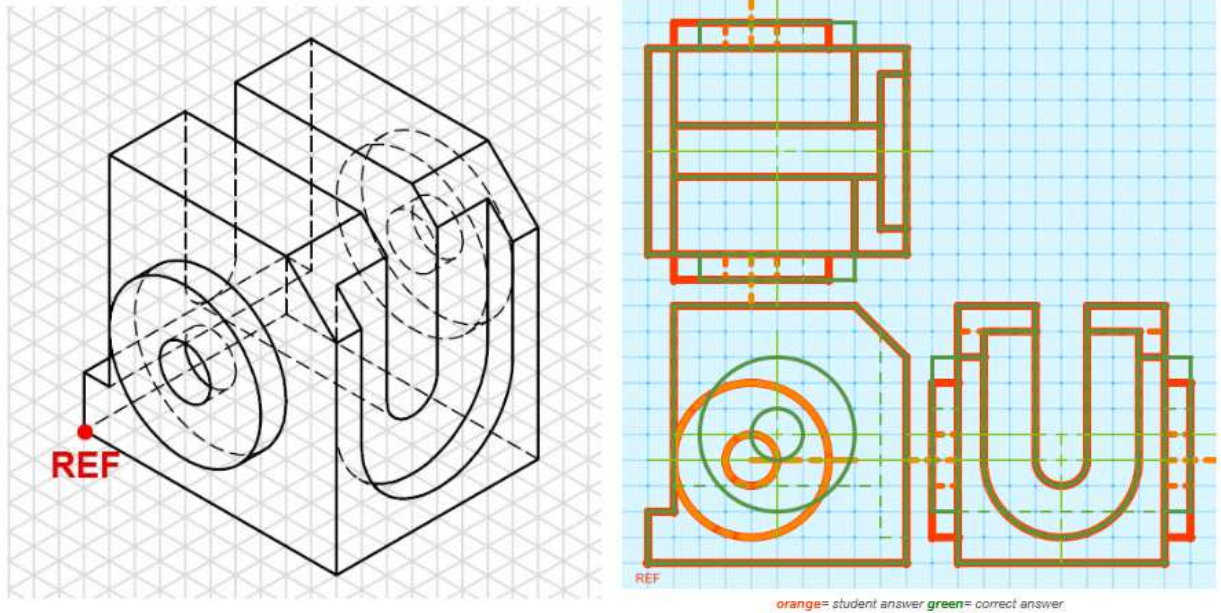
These info graphics let students know exactly how they are doing in real-time i.e. while grade changes are happening asynchronously. Seeing how they are doing compared to other students is rare in many classrooms and is something students frequently ask about.

students can click each box to see more information about that value, and a modal like the one below will display

! ALERT!	
hw01(A)=	59.49
hw02(B)=	50
hw03(C)=	94
hw04(D)=	wip
hw05(E)=	wip
min hw(F)=	0
me01(G)=	wip
fe01(H)=	wip
grade = $0.3 * (((A + B + C + D + E) - F) / 4) + (0.3 * G) + (0.4 * H) = 76.74$	
✖ CLOSE	

8.3.6. On assignments and exams, student can check what they did wrong in detail, like in this assignment example below.

ORTHOGRAPHIC GRADER:



DEDUCTIONS:

$$\text{grade} = \text{cap} @ E(\text{sum}(F)) = \text{cap} @ E(\text{sum}(((A + 0.2 * B) / C) * D) * E)) = 40.51$$

lines that are correct but overlaid or concatenated are not considered mistakes

line type	line deduction(points) (F)	student # of missing entities (A)	student # of excessive entities (B)	total # of possible correct entities (C)	X100% line weight (D)	max point deduction (E)
visible line=	3.3	20	20	218	0.5	60
hidden line=	25.53	33	69	33	0.3	
center line=	11.68	70	15	75	0.2	

RETURN TO GRADE PAGE

8.4. Teacher:

8.4.1. Teacher makes grade adjustments as needed using the student controls from the administration pages.

8.4.2. At the end of semester, teacher either exports grade sheet or views the final grades on the data table viewer below.

PREVIEW TABLE																						
id	studentid	semester	class	session	name	hw01	hw02	hw03	hw04	hw05	me01	fe01	hw06	hw07	hw08	hw09	hw10	me02	fp01	mid	final	
1		summer2020	mce180	a		50		50	50	50	50	50	50	50		50	50	50	50	50	D	D
2		summer2020	mce181	b		50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	D	F
3		summer2020	mce180	a		50	50	50	50	50	50	50	50	50	50	50	50	50	50	50	D	D
4		summer2020	mce181	b		50	50	50	50	50	50	50	91.6822532380688		100	50	50	50	78.2061034132654	50	C	A-
5		summer2020	mce180	a		98.9613808975834	92.5	97	50	50	50	50	50	50	50	50	50	50	50	50	D	B
6		summer2020	mce181	b		50	50	50	50	50	50	50	96.8054470588235		100	50	50	50	100	50	B-	A
7		summer2020	mce181	b		50	50	50	50	50	50	50	50	97.283783698653	99.9393939393939	50	50	50	92.943076325	50	B-	A

8.4.3. Teacher then submits the final grades one by one on the school record.

Personal Data Account Student **Instructor** Applicant Financial Aid

Grading Schedule Class Roster

[Pick another semester](#) [Pick another roster](#) [Export to Excel](#) [Download](#) [Print](#) [Help](#)

Spring 20 - Final Grade Roster - Class Nbr: 3421

Instructor: Izadnegahdar, Daniel C Grade Roster Status: ✓ Posted - 05/11/2020

Class: MCE 180 Computer Aided Engineering I Grade Roster Access: Grade

Section: 50 (33 students) Time period for grade input: 09 Mar '20 8:00 AM - 12 May '20 11:59 PM

Session: 1 Official Time: 15 Jun '20 1:05 PM

[Initiate Grade Change Request](#) Last Action: NONE [Click here for Grade Definitions](#)

CSU ID	Name	Career	Grade Input	Official Grade	Grade Status
		UGRD	B	S	Posted
		UGRD	C+	S	Posted
		UGRD	B+	B+	Posted
		UGRD	A	A	Posted
		UGRD	B-	B-	Posted
		UGRD	A-	A-	Posted
		UGRD	A	A	Posted
		UGRD	04/24/2020	W	Withdrawn
		UGRD	A	A	Posted
		UGRD	B	B	Posted
		UGRD	A	A	Posted
		UGRD	B+	B+	Posted
		UGRD	A-	A-	Posted
		UGRD	A	A	Posted

9. Student Feedback:

9.1.1. Pros:

9.1.1.1. Overall, students have been pleased using this application and many said it was better than Blackboard. They like the real-time feedback, the simplicity, the progression, and how it's organized.

9.1.2. Cons:

9.1.2.1. The grading algorithm on multiple-choice, data-entry, and cartesian drawings has been very accurate, but on pdf drawings and models, it doesn't always catch all the errors. This has been mitigated by going through an on-going 'tuning' process of adjusting constants after each semester.

9.1.2.2. Currently, the server used is not the best quality; it will lag and be down at times. If this scales, a much more reliable server will be required, especially during exam times.

10. Technical:

10.1. Hosting:

- 10.1.1. *Oswald is hosted on the internet as a website(bit.ly/iamoswald) and can be accessed anywhere. There are no separate site installations.*

10.2. Languages:

- 10.2.1. *HTML, CSS, javascript(frontend)*
- 10.2.2. *C#(controller)*
- 10.2.3. *SQL Server(database)*

10.3. Libraries:

- 10.3.1. *processing.js(drawing canvas)*

10.4. Releases:

- 10.4.1. *44 as of 05/19/20*

10.5. Algorithm Innovations:

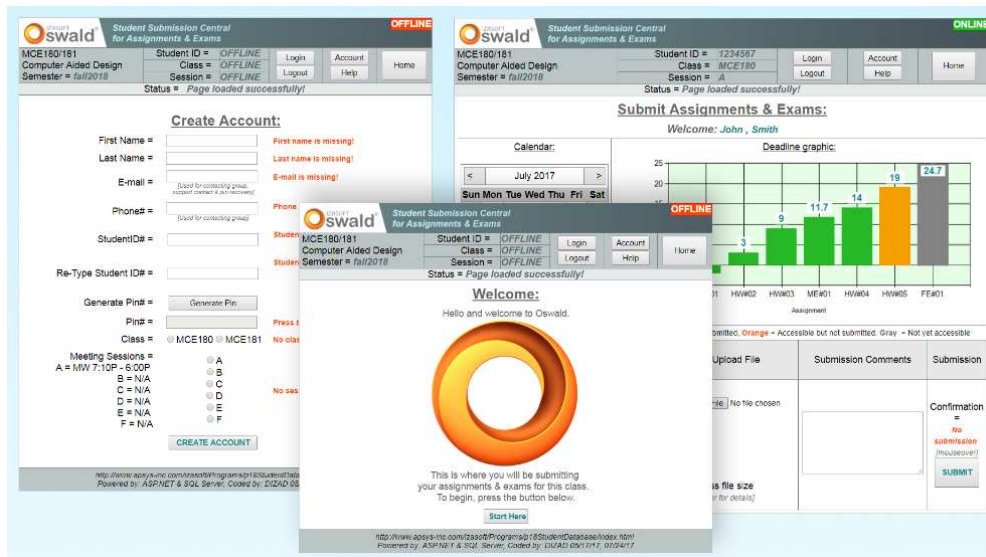
- 10.5.1. *Saving drawings inside SQL tables using custom serialization.*
- 10.5.2. *Grading drawings using a custom cross-referencing atomizer.*
- 10.5.3. *Auto-Scheduling using a custom piggy back algorithm to accommodate calendar fluctuations.*

11. History:

- 11.1. *First year Dan teaches engineering in 2015 where all work was done by 3 teaching assistants, by hand, and hundreds of inconsistencies arose. During this time, Dan was in the early stages of his career change from Mechanical Engineering to Software Engineering and decided to build Oswald.*



11.2. Oswald version 1.0 was released in 2016. It automated 15% of the class and essentially was just a submission hub. It was mainly used to maintain a clean archive and validate file submissions for specific formats and sizes.



11.3. Oswald version 2.0 was released in 2018, and automated all of the remaining book keeping tasks of the class. This was possible through the auto-grading, auto-grouping, and auto-scheduling algorithms.



11.4. *Oswald version 2.0 initial beta test in summer 2018.*



11.5. *Oswald version 2.0 running production in late 2018, running hundreds of algorithms to instantly resolve thousands of hours into seconds.*



12. Conclusion:

- 12.1. *This proposal is an idea to expand Oswald to other classes. Whether this will be successful is an unknown. What is known is that Oswald was successful at significantly reducing the workload of one of the biggest engineering classes at CSU.*
- 12.2. *Most teachers today are still relying on paper and pencil methods to run their class. This is leading to exhaustion, lower pedagogical effectiveness, grading inconsistencies, and higher tuition costs. If teachers knew about the convenience that this technology can provide them, there's a chance that schools would adopt this as a blueprint to their academic pipeline.*